CHAPTER 1

Linux Commands & Terminal Practice

Principles of Linux Operating Systems
Command reference based on Chapter 1

```
Ubuntu terminal | Chapter 1 | practice

updikhadir@ubuntu:~$ whoami
updikhadir
updikhadir@ubuntu:~$ pwd
/home/updikhadir
updikhadir@ubuntu:~$ uname -r
6.x.x-generic
```

How to read the terminal

The prompt `updikhadir@ubuntu:~$` shows the logged-in username, the computer name, the current directory, and the command prompt. The symbol $ indicates a normal user account. In this guide, the username is shown as **updikhadir**, as requested.

Important: Output shown in examples is illustrative. Your Ubuntu version and command results may be different.

1. Basic system information

1. whoami

Displays the username of the account currently running the terminal.

Syntax: `whoami`

```
updikhadir@ubuntu:~$ whoami
updikhadir
```

Example output: updikhadir

2. pwd

Prints the full path of the current working directory. It helps you identify which folder you are currently using.

Syntax: `pwd`

```
updikhadir@ubuntu:~$ pwd
/home/updikhadir
```

Example output: /home/updikhadir

3. uname -r

Displays the release version of the Linux kernel currently running on the computer.

Syntax: `uname -r`

```
updikhadir@ubuntu:~$ uname -r
6.8.0-example-generic
```

Example output: A kernel release string. The exact value depends on the installed system.

4. uname -s

Displays the operating-system kernel name. In the course example, macOS reports Darwin; Linux systems report Linux.

Syntax: `uname -s`

```
updikhadir@ubuntu:~$ uname -s
Linux
```

Example output: Linux

5. cat /etc/os-release

Prints the operating-system identification file. Use it to see the Linux distribution name and version.

Syntax: `cat /etc/os-release`

```
updikhadir@ubuntu:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
```

Example output: The exact fields and version depend on your Ubuntu release.

2. Files and directories

6. ls

Lists files and directories in the current directory. It is one of the basic commands used to inspect folders.

Syntax: `ls`

```
updikhadir@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures
```

Example output: A list of names from the current directory.

7. cp

Copies a file or directory from one location to another. The source chapter names cp as one of the basic shell tools.

Syntax: `cp SOURCE DESTINATION`

```
updikhadir@ubuntu:~$ cp notes.txt notes-backup.txt
```

Example output: No output is normally displayed when the copy succeeds.

8. mv

Moves a file or directory to another location. It can also be used to rename a file.

Syntax: `mv SOURCE DESTINATION`

```
updikhadir@ubuntu:~$ mv draft.txt final.txt
```

Example output: No output is normally displayed when the move or rename succeeds.

3. Package-management commands mentioned in Chapter 1

Linux distributions use different package-management tools. The Chapter 1 material identifies apt for the Debian/Ubuntu family, dnf for the Red Hat family, and zypper for the SUSE family. It also mentions yum as Red Hat-style command language.

9. apt

The package-management tool used on Ubuntu and other Debian-family systems. The chapter identifies apt as the tool used in the lab system.

Syntax: `sudo apt install PACKAGE_NAME`

```
updikhadir@ubuntu:~$ sudo apt install tree
[sudo] password for updikhadir:
Reading package lists...
```

Example output: The package manager reports progress and asks for confirmation when needed.

Note: The package name 'tree' is only an example.

10. dnf

The package-management tool used by Fedora and other systems in the Red Hat family.

Syntax: `sudo dnf install PACKAGE_NAME`

```
updikhadir@ubuntu:~$ sudo dnf install tree
```

Example output: Run this on a distribution that uses dnf, not as the normal Ubuntu package command.

11. zypper

The package-management tool used by openSUSE and SUSE Linux Enterprise.

Syntax: `sudo zypper install PACKAGE_NAME`

```
updikhadir@ubuntu:~$ sudo zypper install tree
```

Example output: Run this on a SUSE-family distribution.

12. yum

An older Red Hat-family package-management command mentioned in the chapter as an example of Red Hat-style instructions.

Syntax: `sudo yum install PACKAGE_NAME`

```
updikhadir@ubuntu:~$ sudo yum install tree
```

Example output: This is not the standard package command for Ubuntu.

4. Shell environment variable: export

The supplied reference image explains **export**. It marks a shell variable so that programs launched from the current shell can read its value. The value normally lasts for the current shell session unless it is added to a shell startup configuration.

13. export NAME=value

Creates or updates a shell variable and exports it to child processes. Use it when a program needs a configuration value through its environment.

Syntax: `export NAME=value`

```
updikhadir@ubuntu:~$ export COURSE=Linux
updikhadir@ubuntu:~$ printenv COURSE
Linux
```

Example output: Linux

14. export with a value containing spaces

Use quotation marks when the value contains spaces so the shell treats it as one value.

Syntax: `export NAME="value with spaces"`

```
updikhadir@ubuntu:~$ export COURSE="Linux basics"
updikhadir@ubuntu:~$ printenv COURSE
Linux basics
```

Example output: Linux basics

15. export -p

Displays exported variables in shell notation.

Syntax: `export -p`

```
updikhadir@ubuntu:~$ export -p
declare -x COURSE="Linux basics"
```

Example output: The full output can contain many exported variables, depending on the session.

Quick revision table

Command	Main purpose	Where it is used
whoami	Shows current username	Basic system information
pwd	Shows current directory path	Navigation / lab
uname -r	Shows kernel release	System information
uname -s	Shows kernel name	OS identification
cat /etc/os-release	Shows distribution details	System information
ls	Lists directory contents	Files and folders
cp	Copies files	File management
mv	Moves or renames files	File management
apt / dnf / zypper / yum	Installs software packages	Depends on distribution

Command	Main purpose	Where it is used
export	Exports environment variables	Shell configuration

Study reminder

Practice each command in your Ubuntu virtual machine. Type one command at a time, press Enter, and compare your output with the examples. Do not copy package-manager commands from another Linux family without checking which distribution you are using.

Source basis: Chapter 1 – Introduction to Linux and Ubuntu (uploaded course slides), plus the supplied export-command reference image. Explanations and examples are concise study notes; example outputs are illustrative.